Emissync

PPT Content for the Internal Project Presentation
22 October 2026 · B.Tech CSE, 7th Semester

How to use this document

- Slides follow the 8 aspects required in the department notice, in the same order.
- Each slide has the **text to put on the slide**, a visual suggestion and a speaker note.
- Items in [brackets] are for you to fill in with your own real data.
- Example numbers assume SWDM v4 at 0.300 kWh/GB and a grid intensity of 442 g CO_2/kWh. Recompute if you use a different grid factor.
- Slide 6 and 7 are based on what the code actually does: network measurement is implemented, the CO_2 conversion is still in progress.

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SLIDE 1 Project Title and Group Members

Emissync

A Real-Time Browser-Level Digital Carbon Monitoring and Attribution System

Real-Time Estimation and Attribution of Carbon Emissions from Web Browsing Sessions

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Supervisor: [Name] | B.Tech CSE, 7th Semester | Internal Project Presentation | 22 October 2026

Tagline: Every Click. Every Byte. One Carbon Impact.

Visual: *clean title slide with a leaf or globe icon and the extension popup screenshot on the right.*

SLIDE 2

Introduction and Background

The Internet has a physical footprint

- The Internet is a global network of networks running on TCP/IP. About 99% of international data travels through 300+ undersea fibre-optic cables.
- ICT (devices, networks, data centres, manufacturing) accounts for an estimated 3.5–3.7% of global greenhouse gas emissions, comparable to aviation (Climate Impact Partners).
- Data centres use about 1% of global electricity. Internet traffic has tripled since 2015, and video and audio streaming make up about 63% of it.

What is a carbon footprint?

- Total direct and indirect GHG emissions across a lifecycle, expressed in **CO_2e** (CO_2, CH_4 and N_2O weighted by global warming potential). *(Univ. of Michigan CSS09-05)*

Two kinds of digital emissions

- **Operational:** electricity used by data centres, networks and user devices during a session.
- **Embodied:** manufacturing, shipping and disposal of hardware. SWDM v4 allocates about 19% of a browsing footprint to it.

Visual: *the IT-sector electricity donut (2017: devices 34%, data centres 29%, networks 21%, manufacturing 16%) next to a simple operational vs embodied diagram.*

Speaker note: *“Every page we load travels through cables, data centres and our own device, and each stage uses electricity. We want to make that visible to the user.”*

SLIDE 3 Problem Statement

The visibility gap

- Web browsing consumes resources across data centres, networks and user devices, but users have no view of the footprint of their own session.
- Existing tools mostly estimate **per URL, from outside the browser**, and report **one number** with no breakdown.
- There is no continuously updated, session-level view that also separates the emission sources.

Research problem

> *How can real-time browser-level measurements be combined with established carbon-estimation methodologies to give a more granular view of the estimated carbon footprint of individual web browsing sessions?*

Key challenge: exact emissions cannot be measured from a browser. They can only be **estimated** from measurable inputs such as bytes, requests and time.

Visual: *a “today vs Emissync” comparison: today is a single opaque number per URL, Emissync is a live, broken-down session view.*

SLIDE 4 Objectives of the Project

1. Build a **Chrome (Manifest V3) extension** that monitors browsing sessions in real time.
2. **Measure** actual network transfer: requests, bytes and resource types, separating cache hits, 304 responses and failures.
3. **Estimate CO_2e** from measured data using an established baseline, **SWDM v4 via CO2.js**.
4. **Attribute** the estimate to data centre, network, user device and embodied hardware.
5. Provide a **live session counter**, a **website carbon profile** and a **history dashboard**.
6. **Compare** the baseline with an Emissync-enhanced estimate (formula below).
7. Keep the work **transparent and privacy-friendly**: local storage only, with results clearly labelled as estimates.

```
Difference (%) = |E_Emissync − E_SWDM| / E_SWDM × 100
```

Speaker note: *Objectives 1–2 are implemented. Objectives 3–6 are the next phases.*

SLIDE 5a Proposed Methodology

5.1 System workflow

```
Browsing activity → webRequest monitor → per-tab request log → classification → bytes by type → carbon engine (SWDM/CO2.js + Emissync layer) → attribution → live dashboard + history
```

5.2 Measurement design (implemented)

- **Source:** Chrome `webRequest` listeners (onBeforeRequest, onCompleted, onErrorOccurred). This captures cross-origin resources, which the Resource Timing API reports as 0 bytes.
- **Per request, record:** URL, type, method, status, cache flag, 304 flag, Content-Length, Content-Encoding, initiator.
- **Classification:** failed | from cache | 304 Not Modified | network. Only real network transfers count toward bytes.
- **Unknown sizes:** responses without Content-Length are flagged, not guessed.
- **Reliable storage:** a serialised queue over `chrome.storage.session` prevents race conditions between concurrent events.
- **Reproducible export:** JSON with a fixed schema and a cold/warm cache label.
- **Privacy:** all data stays local in the browser.

5.3 Calculation

- Baseline: E (kWh) = Data (GB) × 0.300, then $CO_2e = E \times$ grid intensity.
- Attribution: split by SWDM shares (52 / 19 / 15 / 14 %).
- Enhanced model (research layer): add session time and request context, using the time-based device term $E = P \times t$.

Visual: *the workflow as a flow diagram. Put 5.2 and 5.3 on one slide, and 5.4–5.6 on a second "Methodology (contd.)" slide.*

SLIDE 5b

Methodology (contd.): Models, Experiments, Tech Stack

5.4 Estimation models studied

Model	Approach	Role in Emissync
SWDM v4	Bottom-up, 0.300 kWh/GB. Split: device 52%, embodied 19%, data centre 15%, network 14%	**Baseline** (via CO2.js)
SCI, ISO/IEC 21031:2024	SCI = ((E × I) + M) / R per functional unit	Session-level framing (R = one session)
OneByte	Top-down bytes × fixed factors	Comparison only; ignores device energy
EcoIndex	DOM size, request count and payload score	Motivates adding request and DOM context
DIMPACT	Streaming LCA with fixed base load	Shows emissions are not purely linear in data

5.5 Planned experiments

Experiment	Method
A	Measure 20–30 websites: size, session transfer, duration, estimated CO_2e
B	Same websites across browsers on the same machine
C	Page load vs 5-minute vs 10-minute vs media-heavy sessions
D	SWDM baseline vs Emissync-enhanced estimate, % difference

5.6 Tech stack

Chrome Extension (Manifest V3), JavaScript, CO2.js, chrome.storage, Chart.js / Recharts for the dashboard. No AI/ML is required in the core system.

Visual: *the models table and a small diagram of experiments A–D.*

SLIDE 6 Work Completed So Far

Research and design (completed)

- Domain study: internet infrastructure, carbon footprint concepts, operational vs embodied emissions.
- Comparative study of 5 estimation models; **SWDM v4 selected as the baseline**.
- Standards review: RFC 9845, RFC 6988, RFC 8348, IETF GREEN drafts, ISO/IEC 21031:2024.
- System architecture, 7-module design, database schema and experiment design.

Implementation: Phase 1 prototype (v0.1)

- Chrome MV3 extension with network monitoring via `webRequest`.
- Request capture and classification into network, cache, 304 and failed.
- Popup UI: request counts, bytes transferred, unknown-size count, and breakdown by resource type.
- JSON export with a defined data contract and cache-state label, supporting repeatable experiments.
- Serialised storage layer, with automatic cleanup when a tab closes.

In progress

- Integrating the CO2.js/SWDM calculation and showing estimated CO_2e in the popup.

Project status by phase

Phase	Status
1. MVP: measurement	Done (v0.1)
1. MVP: CO_2 estimate in popup	In progress
2. Live dashboard	Planned
3. Attribution	Planned
4. Research layer	Planned
5. Experiments and paper	Planned

Visual: *2–3 popup screenshots on real sites, a short snippet of the exported JSON, and a progress bar for the 5 phases (Phase 1 at roughly 70%).*

Speaker note: *Be direct: “The measurement layer works; the carbon conversion is the next step we are completing now.”*

SLIDE 7 Preliminary Results / Expected Outcomes

Preliminary results (from the prototype)

Measure 3–4 sites with the extension and fill in the table. Convert with **CO_2e (g) = MB × 0.1326** (0.300 kWh/GB × 442 g/kWh ÷ 1000 MB/GB).

Website type	Requests	Transferred (MB)	Unknown-size requests	Est. CO_2e (g)
Text blog	[ ]	[ ]	[ ]	[ ]
News homepage	[ ]	[ ]	[ ]	[ ]
Video site	[ ]	[ ]	[ ]	[ ]

Estimates are calculated offline from measured bytes using SWDM v4.

Observations to highlight (only if your data shows them)

- Media-heavy pages transfer far more data than text pages.
- Images, scripts and video dominate the resource breakdown.
- Cold vs warm cache: a repeat visit transfers far fewer bytes. Show one example.
- Some requests have no size information, which is a known limitation.

Illustrative attribution for a 2 MB page (≈ 0.27 g CO_2e)

Segment	Share	Approx. g CO_2e
User device	52%	0.14
Embodied hardware	19%	0.05
Data centre	15%	0.04
Network	14%	0.04

Expected outcomes

- Working extension with a live estimated CO_2e counter.
- Per-website carbon profile and four-way attribution.
- Daily and weekly history dashboard.
- Quantified difference between the baseline and the enhanced estimate.
- Reproducible methodology and a research paper.

Visual: *a bar chart of MB or g CO_2e per site type and a donut chart of the attribution split.*

Speaker note: *“These are estimates from a documented model, not direct emission measurements.”*

SLIDE 8 Future Work / Work Plan

Phase	Deliverables	Suggested timeline
1. MVP (in progress)	Add CO2.js/SWDM, show estimated CO_2e, auto-refresh popup	Oct 2026
2. Live dashboard	Session timer, live counter, persistent history (storage.local), charts	Nov 2026
3. Attribution	Data centre / network / device / embodied split in the UI	Nov 2026
4. Research layer	Enhanced model and baseline-vs-enhanced % difference	Dec 2026
5. Experiments and paper	Experiments A–D, analysis, limitations, report	Dec 2026 – semester 8

Adjust the dates to your semester calendar and your supervisor's advice.

Planned improvements to the current prototype

- Handle responses with unknown size, for example with fallback estimation.
- Track SPA route changes as separate page views.
- Add the session time dimension for temporal analysis.
- Use a regional grid intensity (for example India) instead of one global value.
- Strip URL query strings before storing or exporting.

Limitations (state them yourself before the panel does)

- Emissync reports **estimated** CO_2e, not exact physical emissions.
- Data transfer is a proxy for resource use, and results depend on model assumptions.
- Content-Length gives compressed body size only, so totals can be undercounted.

References (final slide)

Sustainable Web Design Model and CO2.js (Green Web Foundation) · ISO/IEC 21031:2024 (SCI) · RFC 9845 · RFC 6988 · RFC 8348 · IETF GREEN drafts (Internet-Drafts) · Univ. of Michigan, Center for Sustainable Systems, “Carbon Footprint Factsheet”, Pub. No. CSS09-05 (2026) · Climate Impact Partners, *The carbon footprint of the internet*

PREP

Likely Panel Questions and Short Answers

Question	Short answer
How is this different from existing tools like Carbonalyser or Website Carbon?	Those estimate by URL or by total bytes. Emissync adds live session monitoring, source-wise attribution, temporal analysis and a baseline-vs-enhanced comparison.
How accurate is it?	It is an estimate. We use SWDM, a published model, we flag unknown-size requests, and we compare against the baseline in Experiment D.
Why SWDM as the baseline?	It is bottom-up, segmented, maintained, and built into CO2.js, so results are reproducible.
Why webRequest and not the Performance API?	The Performance API reports 0 bytes for cross-origin resources that lack Timing-Allow-Origin headers.
What is your novelty?	Real-time session measurement, multi-segment attribution, temporal analysis and model validation. Do not claim that nobody has built a carbon extension.
Why no AI/ML?	The core problem is measurement and an established estimation model; AI/ML is not required for it.

Delivery tips

- Aim for about 10–12 minutes, one slide per aspect.
- Show a live demo or screenshots on slide 6 or 7. It is more convincing than text.
- Always say “estimated CO_2e”, never “measured CO_2”.